S

COURSE ASSIGNMENT

AI-Based Early Detection of Alzheimer's Disease Using Medical Images

A Deep-Learning Approach to Classification of Brain MRI Scans

A. Assignment Information

Department	Computer Science and Engineering / Information Technology
Programme	B.Tech CSE
Course Code & Course Name	CSA1708 ARTIFICIAL INTELLIGENCE
Academic Year / Batch	2025-2029
Faculty Name	Dr Jeena
Assignment Title	AI-Based Early Detection of Alzheimer's Disease Using Medical Images
Date of Issue	1/9/2026
Date of Submission	2/9/2026
Maximum Marks	100
Course Outcome(s) – CO	Apply machine learning / deep learning principles to design and evaluate a solution for a real-world engineering problem in medical image analysis
Bloom's Taxonomy Level	L4 (Analyze) / L5 (Evaluate) / L6 (Create)
SDG Mapping (SDG 1–17)	SDG 3 – Good Health and Well-Being; SDG 9 – Industry, Innovation and Infrastructure
Industry / Societal Relevance	AI-assisted diagnostic tools for neurodegenerative disease screening; relevant to healthcare-AI, medical-imaging and diagnostics industry

B. Assignment Problem / Challenge

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder and the most common cause of dementia worldwide, characterized by gradual memory loss, cognitive decline and, eventually, loss of independent function. Because irreversible neuronal damage begins years before clinical symptoms are obvious, early and accurate detection — ideally at the Mild Cognitive Impairment (MCI) stage — is critical for timely intervention, care planning and the evaluation of disease-modifying therapies. Structural and functional neuro-imaging (MRI and PET) captures measurable biomarkers of neurodegeneration, such as hippocampal atrophy and reduced glucose metabolism, but manual interpretation by radiologists is time-consuming, subject to inter-observer variability, and does not scale to population-level screening.

This assignment requires you to formulate, design and analytically justify an Artificial Intelligence / Deep Learning based solution that uses medical brain-imaging data to classify subjects into diagnostic categories (Cognitively Normal, Mild Cognitive Impairment, and Alzheimer's Disease), applying core concepts of image processing, convolutional neural networks, model evaluation and engineering trade-off analysis learned in the course.

Relation to Course Outcomes

This problem requires you to apply concepts of digital image pre-processing, feature extraction, supervised learning, convolutional neural network (CNN) architectures, transfer learning, and performance evaluation metrics — directly addressing the course outcome of designing and evaluating an AI/ML-based solution for a real engineering problem.

C. Problem Statement

Problem / Case / Design Challenge:

Design and analytically justify a Convolutional-Neural-Network-based image classification system that takes as input 2D/3D brain MRI (or PET) scan slices and predicts the subject's cognitive status — Cognitively Normal (CN), Mild Cognitive Impairment (MCI), or Alzheimer's Disease (AD) — with an associated confidence score. The system should be developed using a publicly available, de-identified neuro-imaging dataset (e.g., the ADNI — Alzheimer's Disease Neuroimaging Initiative — dataset, or the Kaggle 'Alzheimer's MRI Preprocessed Dataset'), and its performance must be validated using standard classification metrics (accuracy, precision, recall, F1-score, confusion matrix). You are expected to compare at least two candidate modelling approaches (e.g., a CNN trained from scratch versus a transfer-learning model such as VGG16/ResNet50), justify the chosen final approach with quantitative evidence, and discuss the broader ethical, safety and societal implications of deploying such a system as a diagnostic-support tool.

D. Requirements and Constraints

The following functional, technical, ethical and resource-related requirements and constraints are relevant to this problem and must be addressed in the solution:

Category	Requirement / Constraint
Functional requirement	The system shall classify a brain MRI (or PET) scan into diagnostic categories — Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD) — and output a class probability/confidence score.
Performance requirement	Target classification accuracy of at least 85–90% on a held-out test set, with sensitivity (recall for the AD class) prioritized to minimize missed early-stage cases.
Technical constraint	Limited availability of large, labelled, balanced neuro-imaging datasets; high computational cost of training deep 3D CNNs; variability in scanner protocols and image resolution across sites.

Category	Requirement / Constraint
Data constraint	Use of a public, de-identified dataset (e.g., ADNI or the Kaggle Alzheimer's MRI dataset) since access to real patient data requires institutional ethics approval.
Cost / Time	Solution must be trainable within the compute budget of a free-tier GPU environment (e.g., Google Colab) and within the assignment timeline.
Reliability	Model predictions must be reproducible across runs (fixed random seeds) and robust to minor variations in image contrast, orientation and noise.
Safety / Ethical considerations	The tool is a decision-support aid, not a replacement for a radiologist/clinician; false negatives (missed AD cases) are more critical than false positives and must be explicitly discussed.
Security / Privacy	Patient scans and metadata must be de-identified; any deployment must comply with data-protection norms (e.g., HIPAA/GDPR-equivalent handling).
Standards	Where applicable, DICOM image standards for medical imaging and standard train/validation/test splitting protocols must be followed.
User requirement	Output must be interpretable to a non-expert reviewer — e.g., via a class-activation/Grad-CAM heatmap highlighting the brain regions that influenced the prediction.

E. Student Work

1. Problem Understanding and Formulation

What is the problem?

The core problem is a multi-class image classification task: given a brain MRI scan, automatically determine whether the subject shows no cognitive impairment (CN), early-stage impairment (MCI), or Alzheimer's Disease (AD). This is fundamentally a pattern-recognition problem in which subtle structural differences — such as hippocampal and cortical atrophy, ventricular enlargement, and grey-matter density loss — must be learned from labelled image data rather than encoded through hand-crafted rules, because these patterns are too complex and variable across individuals for a simple algorithmic rule.

What are the expected outcomes?

- A trained deep-learning classification model that maps an input MRI slice/volume to one of the CN / MCI / AD classes.
- Quantitative performance evaluation of the model (accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC).
- A comparative analysis of at least two modelling approaches with a justified final recommendation.
- A visual explanation mechanism (e.g., Grad-CAM) so predictions are interpretable rather than a 'black box'.

What information / data is available?

Publicly available, de-identified neuro-imaging datasets are available for this task, most notably the ADNI (Alzheimer's Disease Neuroimaging Initiative) dataset, which provides longitudinal MRI, PET, clinical and cognitive-assessment data for CN, MCI and AD subjects, and the Kaggle 'Alzheimer's MRI Preprocessed Dataset', which offers pre-sliced, labelled 2D axial MRI images across four severity categories (Non-Demented, Very Mild, Mild, Moderate Demented) and is well suited to an assignment-scale project given limited compute resources.

What assumptions are made?

- The provided image labels (diagnostic category) are clinically validated and reasonably accurate ground truth.
- A 2D slice-based or pre-processed dataset is an acceptable simplification of full 3D volumetric analysis for the scope of this assignment.
- Class imbalance (typically fewer AD/MCI samples than CN) will be handled through data augmentation or class-weighting rather than collecting new data.
- The trained model is a decision-support prototype, not a certified clinical diagnostic device.

What constraints must be satisfied?

The solution must satisfy the functional, performance, technical, data, ethical and resource constraints listed in Section D — in particular, it must run within limited GPU compute, use only public de-identified data, and prioritize minimizing false negatives for the AD class.

2. Application of Course Knowledge

This solution draws on the following principles, theories and mathematical models from digital image processing, machine learning and deep learning:

2.1 Image Pre-processing

- Intensity normalization — rescaling pixel intensities to a standard range (e.g., 0–1) so that the network is not biased by scanner-specific intensity variation.
- Skull-stripping / brain extraction — removing non-brain tissue so the model focuses on relevant anatomical structures.
- Resizing and resampling to a fixed input resolution (e.g., 128×128 or 224×224 pixels) for batch processing.
- Data augmentation (rotation, horizontal flip, zoom, small translation) to artificially increase training-set diversity and reduce overfitting, given the limited number of AD/MCI samples.

2.2 Convolutional Neural Network (CNN) Fundamentals

A CNN learns spatial hierarchies of features through the convolution operation. For an input image I and kernel K, the 2-D discrete convolution at output position (i, j) is:

$$S(i, j) = \Sigma m\ \Sigma n\ I(i - m, j - n) \cdot K(m, n)$$

Each convolutional layer is typically followed by a non-linear activation (ReLU: $f(x) = \max(0, x)$) and a pooling operation (e.g., max-pooling) that progressively reduces spatial dimensionality while retaining the most salient activations. Stacking several such layers allows the network to learn low-level features (edges, textures) in early layers and high-level, disease-relevant structural patterns (e.g., hippocampal shrinkage) in deeper layers.

2.3 Loss Function and Optimization

For multi-class classification, the categorical cross-entropy loss is used:

$$L = - \Sigma c{=}1..C\ \ yc \cdot \log(\hat{y}c)$$

where C is the number of classes, yc is the true one-hot label and ŷc is the predicted probability for class c (obtained via a final softmax layer). The network weights are updated using a gradient-based optimizer such as Adam, which combines momentum and adaptive per-parameter learning rates to accelerate and stabilize convergence.

2.4 Transfer Learning

Because labelled medical-imaging data is scarce, transfer learning is applied: a CNN (e.g., VGG16, ResNet50 or InceptionV3) pre-trained on a large natural-image dataset (ImageNet) is reused, with its convolutional base frozen (or fine-tuned) and its final classification head replaced/retrained for the CN/MCI/AD task. This leverages generic low-level visual features learned from millions of natural images and adapts only the higher-level decision layers to the medical domain, substantially reducing the amount of labelled data and training time required.

2.5 Evaluation Metrics

Accuracy = (TP + TN) / (TP + TN + FP + FN); Precision = TP / (TP + FP); Recall (Sensitivity) = TP / (TP + FN); F1-score = 2 · (Precision · Recall) / (Precision + Recall). In this application, Recall for the AD and MCI classes is prioritized over raw accuracy, because a false negative (missing an actual AD/MCI case) carries a higher clinical cost than a false positive.

3. Solution / Design / Methodology

3.1 Overall Pipeline

The proposed solution follows a standard supervised deep-learning pipeline:

1. Data acquisition: obtain a labelled MRI dataset (ADNI or Kaggle Alzheimer's MRI dataset).
2. Pre-processing: normalization, skull-stripping (if using raw ADNI volumes), resizing, and train/validation/test split (e.g., 70% / 15% / 15%), stratified by class.
3. Data augmentation: applied only to the training set to mitigate class imbalance and overfitting.
4. Model design: two candidate architectures are built for comparison — (a) a custom CNN trained from scratch, and (b) a transfer-learning model using a pre-trained backbone.
5. Training: both models are trained with categorical cross-entropy loss, the Adam optimizer, early stopping (monitoring validation loss) and dropout regularization to reduce overfitting.
6. Evaluation: both models are evaluated on the held-out test set using accuracy, precision, recall, F1-score and confusion matrix.
7. Interpretability: Grad-CAM is applied to the better-performing model to visualize the brain regions driving each prediction.
8. Comparison and selection: the two approaches are compared quantitatively and qualitatively, and a final recommended model is justified.

3.2 Candidate Architecture A — Custom CNN (baseline, trained from scratch)

- Input: 128×128×1 (grayscale) pre-processed MRI slice.
- Conv Block 1: Conv2D(32 filters, 3×3) → ReLU → MaxPooling(2×2).
- Conv Block 2: Conv2D(64 filters, 3×3) → ReLU → MaxPooling(2×2).
- Conv Block 3: Conv2D(128 filters, 3×3) → ReLU → MaxPooling(2×2).
- Flatten → Dense(128, ReLU) → Dropout(0.5) → Dense(3, Softmax) for CN / MCI / AD.

3.3 Candidate Architecture B — Transfer Learning (VGG16 / ResNet50 backbone)

- Input: 224×224×3 (RGB-replicated) pre-processed MRI slice, matching the pre-trained backbone's expected input.
- Frozen convolutional base pre-trained on ImageNet (weights not updated in the first training phase).
- Global Average Pooling layer to reduce the feature-map output to a fixed-length vector.
- New classification head: Dense(256, ReLU) → Dropout(0.5) → Dense(3, Softmax).
- Optional fine-tuning phase: unfreeze the last few convolutional blocks and continue training at a lower learning rate to adapt higher-level features to the medical-imaging domain.

Both candidates are trained under identical data splits, augmentation and evaluation protocols so that the comparison in Section 6 isolates the effect of architecture/transfer learning rather than confounding factors.

4. Use of Modern Tools

The following modern engineering/computing tools are appropriate for implementing and evaluating this solution:

- Python 3 — primary implementation language.
- TensorFlow / Keras (or PyTorch) — for building, training and fine-tuning the CNN and transfer-learning models.
- Google Colab / Jupyter Notebook — free-tier GPU-accelerated environment suitable for assignment-scale training.
- NumPy, Pandas — numerical computation and dataset/label management.
- OpenCV / Pillow, scikit-image — image loading, resizing, normalization and skull-stripping pre-processing.
- scikit-learn — computing evaluation metrics (precision, recall, F1, confusion matrix) and train/test splitting.
- Matplotlib / Seaborn — plotting training/validation accuracy-loss curves, confusion matrix heatmaps and ROC curves.
- Grad-CAM (via tf-keras-vis or a custom implementation) — visual explanation of model predictions.
- Git / GitHub — version control of the notebook/code and reproducibility of the submitted work.

Students should include evidence of tool usage in the actual submission — e.g., annotated code screenshots, notebook cell outputs, or a link to the executed notebook/repository.

5. Results and Validation

The following illustrative results demonstrate the expected format and type of evidence to be presented (students must replace these with their own experimentally obtained values from their trained models):

5.1 Illustrative Confusion Matrix (Transfer-Learning Model, Test Set, n = 440)

Actual \ Predicted	CN	MCI	AD
CN	142	9	2
MCI	11	118	14
AD	1	12	131

5.2 Illustrative Performance Metrics

Class	Precision	Recall	F1-score
CN	0.92	0.93	0.925
MCI	0.85	0.83	0.84
AD	0.89	0.91	0.90
Overall Accuracy	—	—	0.887 (88.7%)

5.3 Additional Evidence to Include

- Training vs. validation accuracy and loss curves across epochs, to demonstrate whether the model is under-fitting, well-fitted, or overfitting.
- ROC curves and AUC values per class for a threshold-independent view of separability between CN, MCI and AD.
- Grad-CAM heatmap overlays on sample MRI slices, showing that the model attends to clinically plausible regions (e.g., hippocampus, medial temporal lobe) rather than irrelevant background.

5.4 Validation Against Requirements

The illustrative results show an overall accuracy of 88.7%, which satisfies the target performance requirement of 85–90% accuracy stated in Section D. Recall for the AD class (0.91) exceeds recall for the other classes, satisfying the requirement to prioritize sensitivity for Alzheimer's detection and thereby minimizing missed true-positive AD cases. The confusion matrix shows that most mis-classification occurs between the CN and MCI classes — expected, given the subtlety of early-stage structural change — rather than between CN and AD, which is clinically reassuring since gross CN/AD confusion would be the most serious type of error.

6. Analysis and Engineering Decision

6.1 Comparison of Candidate Approaches

- The custom CNN trained from scratch (Architecture A) typically achieves lower accuracy (illustratively ~78–82%) because the limited dataset size is insufficient to learn robust low-level filters from random initialization, and it is more prone to overfitting despite dropout and augmentation.
- The transfer-learning model (Architecture B) generally achieves higher accuracy (illustratively ~87–90%) and converges in fewer epochs, because it reuses rich, general-purpose visual features already learned from a much larger natural-image corpus, requiring the network to learn only the mapping from those features to the CN/MCI/AD decision.
- Training time and computational cost: Architecture A trains faster per epoch (smaller network) but needs many more epochs to converge; Architecture B's frozen-base phase trains quickly, while the optional fine-tuning phase is more computationally expensive per epoch.
- Interpretability: both architectures support Grad-CAM visualization, but the deeper pre-trained backbone in Architecture B tends to produce more spatially coherent activation maps.

6.2 Trade-offs

The transfer-learning approach trades a larger memory/parameter footprint and dependency on a pre-trained backbone (requiring internet access or a bundled weights file) for substantially better accuracy and faster convergence on a small medical dataset. The custom CNN trades lower accuracy for a smaller, fully self-contained model that may be preferable in a constrained deployment environment (e.g., limited storage) or where full control over every learned filter is required for regulatory transparency.

6.3 Final Engineering Decision

Based on the quantitative comparison, the transfer-learning model (fine-tuned ResNet50/VGG16) is recommended as the final solution because it meets the target accuracy and recall requirements from Section D with a smaller labelled dataset and shorter training time than training a comparably deep CNN from scratch — directly addressing the technical constraint of limited data availability identified in Section D. This decision should be re-validated against a chosen team's actual experimental results rather than the illustrative figures used here.

7. Broader Considerations

- Sustainability / Environment: training deep-learning models consumes energy; using transfer learning and early stopping reduces the number of training epochs and hence the computational carbon footprint compared to training very deep networks from scratch.
- Society: an accurate, accessible screening aid could extend early-detection capability to under-resourced clinics lacking specialist neuro-radiologists, but must not be presented as a replacement for clinical judgement.
- Safety: false negatives (missed AD/MCI cases) delay treatment and care planning; the system must be positioned strictly as a decision-support tool with clear disclaimers, and any deployment would require clinical validation and regulatory approval (e.g., as a Software as a Medical Device).
- Ethics: informed consent and data-protection compliance are mandatory for any real patient data; algorithmic bias must be checked, since a model trained predominantly on one demographic/scanner population may generalize poorly to others.

- Accessibility: a lightweight, transfer-learning-based model can plausibly run on modest hardware, improving accessibility for smaller healthcare facilities compared to solutions demanding specialized infrastructure.
- Professional responsibility: engineers and developers of such tools are responsible for transparent reporting of model limitations, uncertainty, and appropriate use, and must involve clinical experts in validation before any real-world use.

8. Conclusion

This assignment formulated the early detection of Alzheimer's Disease from brain MRI as a supervised multi-class image-classification problem and proposed a deep-learning pipeline combining pre-processing, data augmentation, and two candidate architectures — a custom CNN and a transfer-learning model — evaluated using standard classification metrics. The transfer-learning approach was found to better satisfy the target accuracy and sensitivity requirements given the constraint of limited labelled medical data, and Grad-CAM visualization was proposed to make the model's predictions clinically interpretable. Limitations include reliance on 2D slices rather than full 3D volumetric data, use of a relatively small public dataset that may not generalize across scanner types and populations, and the prototype's status as a decision-support aid rather than a validated clinical tool. Possible improvements include using 3D CNNs on full MRI volumes, incorporating multi-modal data (MRI + PET + clinical/cognitive scores), applying cross-site external validation, and conducting a formal clinical trial with expert-radiologist-labelled ground truth.

9. Student Reflection

What did you learn from this assignment beyond what was directly taught in the classroom?

Draft starting point: Beyond the classroom coverage of CNN theory, this assignment required understanding domain-specific challenges of medical imaging — such as data scarcity, class imbalance, the clinical cost-asymmetry between false negatives and false positives, and the practical value of transfer learning when labelled data is limited. It also required translating an ambiguous, open-ended healthcare problem into a well-defined, testable engineering formulation, and considering interpretability and ethics as first-class design requirements rather than an afterthought.

If you were given additional time/resources, what would you improve and why?

Draft starting point: With more time and compute, I would move from 2D slice-based classification to full 3D volumetric CNNs, which better capture spatial relationships across the entire brain; incorporate additional modalities such as PET imaging and cognitive test scores in a multi-modal fusion model; and validate the model on an independent, multi-site dataset to test generalizability across different scanners and populations before considering any clinical relevance.

10. References

(Use a consistent citation style, e.g., IEEE or APA, as specified by your faculty. Update this list with the exact sources you actually use.)

- Alzheimer's Disease Neuroimaging Initiative (ADNI). Publicly available longitudinal MRI/PET/clinical dataset for Alzheimer's research. Available: https://adni.loni.usc.edu
- Kaggle. "Alzheimer's MRI Preprocessed Dataset." Available: https://www.kaggle.com/datasets
- Simonyan, K., & Zisserman, A. (2014). Very Deep Convolutional Networks for Large-Scale Image Recognition (VGG). arXiv preprint.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition (ResNet). Proceedings of CVPR.
- Selvaraju, R. R., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. Proceedings of ICCV.
- Chollet, F. (2021). Deep Learning with Python (2nd ed.). Manning Publications.

- TensorFlow / Keras official documentation. Available: https://www.tensorflow.org
- scikit-learn official documentation. Available: https://scikit-learn.org
- [Add any AI-assisted tools used in preparing this assignment, per your institution's academic-integrity / AI-use disclosure policy.]

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment; the mapping below is indicative and should be reviewed/adjusted by the course faculty.

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO5	1,2,3,4,5,8,9,11, 12	L3/L4	
Application of knowledge	CO5	1,2,3,4,5,8,9,11, 12	L3/L4	
Solution / Design / Implementation	CO5	1,2,3,4,5,8,9,11, 12	L4/L5/L6	
Modern tool usage	CO5	1,2,3,4,5,8,9,11, 12	L3/L4	
Validation	CO5	1,2,3,4,5,8,9,11, 12	L4/L5	
Analysis & justification	CO5	1,2,3,4,5,8,9,11, 12	L4/L5	
Broader considerations	CO5	1,2,3,4,5,8,9,11, 12	L4/L5	
Documentation & reflection	CO5	1,2,3,4,5,8,9,11, 12	L3/L4	

H. Faculty Evaluation & Continuous Improvement

tCO Attainment

Performance Level	Criterion
Level 3	≥ 70%
Level 2	60–69%
Level 1	50–59%
Level 0	< 50%

Target CO Attainment: ≥70%

Actual CO Attainment: 87

Faculty Analysis

Areas in which students performed well: Actively participated in class activities.

Areas requiring improvement: Needs better understanding of advanced topics.

Corrective / Improvement Action: Conduct revision and doubt-clearing sessions.

Action to be implemented in the next offering: Give more examples for advanced concepts.